

Dispatch Cost and Mechanism Activation in Wireless Chunk Scheduling

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Abstract—Chunking amortizes scheduling overhead but can increase tail imbalance and exposure to worker loss. We evaluate this tradeoff in an instrumented heterogeneous wireless model using a calibrated ns-3-derived radio lookup, matched scheduling variants, and common recovery. A serialized coordinator-service sweep retunes fixed chunk size independently at every cost level. Core and mechanism-activation cells use 300 paired evaluation seeds; cost-sweep points use 200. Independent delay, continuously varying eligible links, and compute-heavy departures exercise mechanisms that are inactive in simple cells. Paired Student inference supplies simultaneous intervals and difference tests, with explicit equivalence testing. All scenarios use a calibrated stationary radio lookup and common corrected recovery. In the combined zero-service-cost scenario, the full policy has signed gain -4.65% against separately tuned fixed-size dispatch. The first tested service cost with a practical win over single-task dispatch is 5000 ms. Against retuned fixed chunks, signed gains are 1.26% at 5000 ms and -32.10% at 20000 ms. Comparisons with a retuned comparator, independent-delay and useful-link activation cells, and equivalence tests limit the interpretation of these effects. Coordinator costs and injected delays are unmeasured sensitivity settings; the evidence does not establish physical-device gains or a universally superior scheduler.

Index Terms—Wireless computing, heterogeneous scheduling, prefetch, task recovery, simulation validity, sensitivity analysis.

I. INTRODUCTION

A shrinking allocation can reduce the number of tasks stranded on a slow worker near job completion. Prefetch can overlap input transfer and computation, but can also reserve tasks on a worker that subsequently becomes unavailable. Link-quality factors aim to limit exposure to poor communication. These mechanisms interact with channel modeling and recovery: a purported scheduler advantage can instead reflect an unusually pessimistic rate mapping or bandwidth consumed by obsolete assignments.

This study asks three questions. First, do shrinking and link factors outperform a calibrated fixed-size comparator when communication uses a packet-simulator-derived lookup? Second, do restricting prefetch to stationary workers and stopping it at a completion threshold improve performance? Third, do added coordinator cost and independently varying delay change those comparisons? The contribution is a dispatch-cost and activation analysis: it identifies when amortization can offset fine-grained dispatch, while distinguishing superiority over a single-task policy from superiority over a retuned comparator.

The experiments combine four synthetic scenarios, matched mechanism removals, independent comparator tuning, and a

coordinator-cost sweep. The main contrasts use the same independent channel lookup and common runtime, with distinct assumptions stated for coordinator costs and injected delays. Completion and horizon-restricted time are reported together. Physical mobility and end-to-end network execution remain outside the inference scope.

II. RELATED WORK AND COMPARISON SCOPE

Guided self-scheduling and factoring reduce allocation granularity as work decreases [1], [2]. Trapezoid self-scheduling reduces chunk sizes according to a different progression [3]; adaptive weighted factoring responds to changing worker capability [4]. Modern dynamic loop-scheduling libraries expose related tradeoffs on heterogeneous processors [5]; chunk reshaping is also studied in heterogeneous data-processing clusters [6]. Consequently, decreasing chunks and capability weighting alone do not establish algorithmic novelty.

HEFT uses task ranking and earliest predicted finish to schedule heterogeneous resources [7]. HEFT addresses a different ranking/EFT problem; no complete HEFT runtime is used as a primary performance comparator here. The main comparisons instead isolate mechanisms in one common instrumented runtime, which permits a narrower causal interpretation.

Mobile crowd-computing and offloading frameworks address deployment-level questions beyond chunk allocation [8], [9], [10], [11]. This study does not reproduce these systems and makes no comparative performance claim against them. Work stealing addresses decentralization and utilization [12]; LATE and Dolly use speculative execution and cloning to mitigate stragglers [13], [14]. MAUI and CloneCloud illustrate mobile code-offload designs with additional runtime and energy objectives [15], [16]. These systems supply related mechanisms, not performance baselines reproduced here. Granularity decisions trade scheduling/communication overhead against idle time, motivating explicit cost sensitivity rather than a zero-service-cost ranking.

The MAC analysis of Bianchi [17] and the 802.11 performance anomaly [18] illustrate why a fluid rate table cannot by itself validate contention. ns-3 provides an independent packet-level modeling framework [19], but agreement between two simulators does not establish real-world validity. Optional drift-detection code uses ADWIN [20]; drift detection is excluded from the evaluated policy and is not a contribution claimed here.

III. MODEL AND CHANNEL CALIBRATION

A. Tasks, Workers, and Participation

A coordinator distributes independent indivisible task identifiers to eight workers. Each job has 240 tasks with nominal compute cost averaging ten reference-worker seconds. Compute heterogeneity spans a factor of four while preserving total nominal speed. Input size is proportional to task cost; results contain one tenth of the input volume. Nominal task costs are known, while worker capability is estimated from completed compute intervals. These assumptions favor instrumented workloads and do not cover unknown-size jobs or DAG dependencies.

The scenarios are uniform links, a stationary weak-link group, intermittent participation, and combined weak links, participation changes, and compute slowdowns. Half of the workers belong to the weak group where applicable; half have dynamic participation where applicable. Availability events include temporary loss, permanent departure, and departure followed by return. Signal traces include synthetic correlated fluctuations and dips. The main cells control signal groups directly; geometric random-walk coordinates do not independently validate their radio traces. Stationary designation is supplied to the scheduler. Full parameters are provided in the supplement.

B. Independent Packet-Level Calibration

ns-3 3.44 models an IEEE 802.11a AP and stationary receivers, 20 MHz at 5 GHz, ideal rate adaptation, 16 dBm transmit power, and log-distance loss with exponent three and reference loss 56 dB. Saturated UDP downlink application goodput is observed for five seconds after association; separate trials send 100 echo probes. AP data-frame RSSI excludes neighboring-station ACKs. With no decoded frame, the lookup coordinate is configured receive power, not an observed RSSI. Echo failure remains a failure: it is represented by a declared 400 ms probe timeout, not zero delay.

Training RNG runs 1–3 fit piecewise-linear goodput and conditional echo-delay knots. Runs 4–5 evaluate the same positions. Initial diagnostics also vary receiver count; the expanded low-signal support is single-receiver calibration. There are 190 packet-level runs in total. Holdout at training distances tests stochastic repeatability, not unseen-distance accuracy. No fading, measured interference, TCP, or mobile scheduler is calibrated.

The expanded single-receiver holdout goodput MAE is 0.015 Mbps. Small error at identical stationary training positions reflects limited stochastic variation and does not validate moving or heterogeneous contending stations.

C. Projection into the Scheduler

Every core scenario uses the expanded goodput lookup for data transfer and the training conditional echo-delay lookup for observations. Out-of-support signals are rejected rather than silently extrapolated. Synthetic departure state is combined with nonzero lookup goodput and probe delivery. An active transfer receives its nominal application-goodput rate

divided by the number of active positive-rate transfers. This equal-airtime projection is assumed; it is not a heterogeneous moving-station MAC calibration. Conditional unloaded echo does not reproduce queueing delay under data load. Probes and compute-progress reports remain instantaneous. Per-allocation coordinator service and dispatch-control volume are explicitly charged in the cost and activation experiments.

Thus, the primary experiment recalibrates all core rate and echo mappings while retaining explicit approximations to contention and mobility. It is neither a fully calibrated network simulator nor an end-to-end scheduler executed inside ns-3. The lookup alone cannot establish measured mobile-cluster gains.

IV. SCHEDULING AND RECOVERY MECHANISMS

Let R denote unassigned task count and U eligible workers. Compute-only EWMA speed is $\hat{s}_i$, initialized equally and updated from local compute duration. The link-aware chunk rule is

$$k_i = \min \left(R, \max \left(1, \left\lceil 3R \frac{\hat{s}_i}{\sum_{j \in U} \hat{s}_j} \ell_i \delta_i \right\rceil \right) \right), \quad (1)$$

with $\ell_i = \text{clip}((\text{RSSI}_i + 85)/25, .25, 1)$ and $\delta_i = \text{clip}(20/\text{RTT}_i, .25, 1)$, using RTT in milliseconds. A matched compute-only policy removes both factors and prefetch while retaining shrinking and recovery. Removing only RSSI or RTT preserves the other factor and the common eligibility test. Small chunks, clipping, and the critical completion path can make a removal change nothing even when the formula differs.

Idle workers have priority over prefetch. The full policy reserves at most one next chunk for a stationary worker while its current chunk computes. Estimated remaining compute must exceed nominal input-transfer duration plus one second. Prefetch stops at 75% unique accepted completion; existing reservations are retained. These settings are hypotheses under test, not established optimal values. An independent-delay cell adds 80 ms to every conditional echo delay; a continuous-link cell varies half the workers between -78 and -72 dBm, always above eligibility; a compute-heavy departure cell reduces input/compute ratio to 0.01. The common 80 ms offset places the RTT multiplier at its lower clip; this tests the effect of that scaling regime rather than delay-based ranking among workers. These are declared synthetic interventions, not measured load traces. The unrestricted-worker and no-cutoff variants remove the corresponding restriction.

Eligibility requires reachability, RSSI at least -80 dBm, and observed RTT at most 150 ms. Three unhealthy probes, four seconds apart, start a 20-second grace period. Expiry advances the session epoch and returns unaccepted task IDs to the pending set. Task-ID deduplication and epoch fencing reject repeated and stale results. The RSSI threshold is an assumed participation rule, not a validated physical-departure detector.

In the primary runtime, expiry also cancels obsolete coordinator transfers. A transfer waiter receives cancellation and cannot later commit its result. Local computation already in progress can continue during disconnection; a stale output

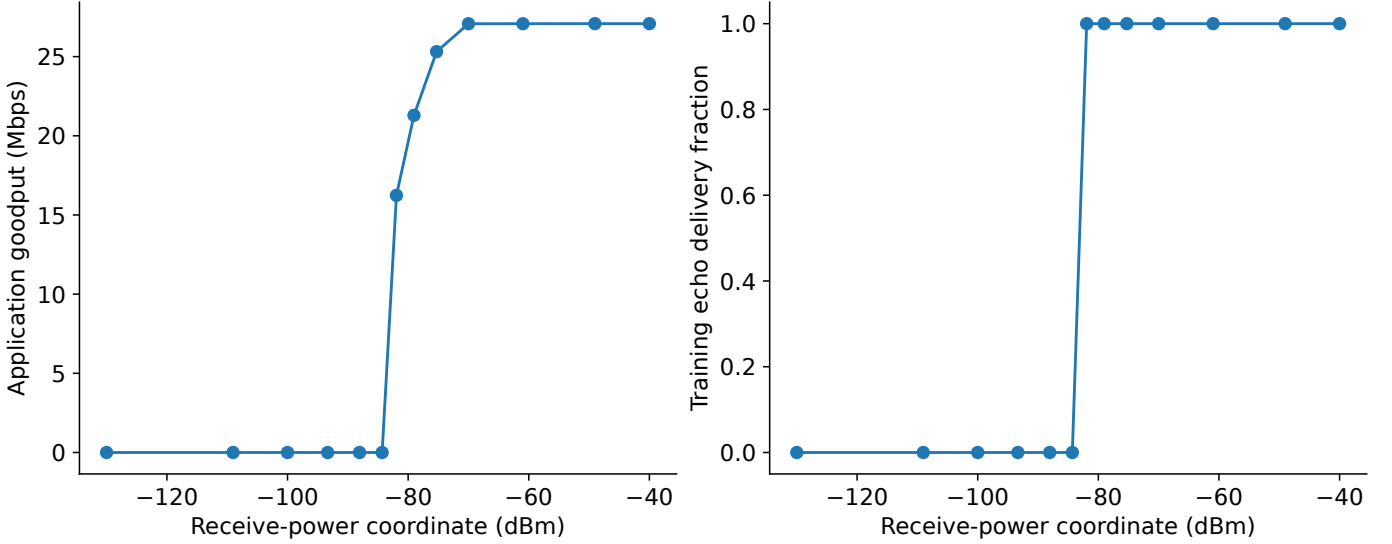


Fig. 1. Expanded ns-3-derived application-goodput lookup and echo delivery. Configured receive power is used where no data frame is decoded. The missing-echo timeout is a protocol convention, not a latency measurement.

is suppressed. All schedulers share this corrected recovery. Transfer retention is a simulator-correctness lesson, not a claimed comparison against real systems that deliberately leave dead sockets transmitting.

V. EXPERIMENTAL AND STATISTICAL PROTOCOL

A. Dispatch Service and Control Cost

Each allocation requests a serialized coordinator resource for a fixed service interval. Its combined request/reply control budget is then represented as one transfer through the same fluid channel as data. Costs are charged once per allocation, including prefetch and retries. Core cases use zero added service and control cost; sweep and activation cases use 64 kB of dispatch-control volume. The sweep covers 0,1,10,100,1000,5000,20000 ms of serialized service. Activation cells use 100 ms. Service denotes coordinator processing, not a network RTT that could overlap across workers. Dispatch still polls at one-second intervals, so zero added service does not imply continuous or instantaneous event-driven allocation.

These values are unmeasured sensitivity settings. The upper range stresses coordinator saturation and must not be represented as typical production overhead. The radio trace and workload are identical across sweep costs for each seed; only runtime costs change. Sweep horizon is 10,000 seconds, compared with 4,000 seconds for core/activation cells. Every comparison uses a common horizon within its cell.

B. Tuning, Precision, and Pairing

Fixed sizes 1,2,3,4,8,16,32 are evaluated on 20 independent calibration seeds per cell. Selection minimizes mean horizon-restricted completion time, with ties resolved by smaller size. Each cost point is retuned separately. A one-task baseline is the feasible lower boundary for indivisible tasks and remains visible beside the selected comparator.

Core and activation cells use a fixed budget of 300 independent evaluation seeds, with achieved uncertainty reported

instead of significance-based stopping or guaranteed power. Sweep points use 200 paired worlds. Total evaluation and tuning counts are 22400 and 1960. Calibration seeds are excluded from evaluation. All schedulers use immutable shared worlds and common reliability semantics. Source checks total 46; parameters, complete seed lists and hashes accompany the data. Results remain conditional on synthetic worlds rather than a random sample of physical deployments.

C. Coherent Difference and Equivalence Inference

For seed j , the relative paired effect is

$$d_j = 100 \frac{T_{B,j}^H - T_{A,j}^H}{T_{B,j}^H}, \quad T^H = \min(T, H). \quad (2)$$

The estimand is the mean of seed-level relative effects, rather than a ratio of marginal means. Every seed enters, and completion is separate. Paired Student intervals and tests use the standard error of these independent seed-level effects. The approximation relies on adequate sampling of bounded simulator outcomes; it does not assume task-level independence.

The joint family has 154 hypotheses: a zero-mean difference test and a TOST equivalence test for each of 77 contrasts. Bonferroni correction uses this complete family. Simultaneous difference intervals use the same Student procedure and correction as the difference p values; exclusion of zero therefore agrees with the adjusted difference test. This conservative choice accommodates correlation across reused worlds and costs.

Equivalence requires adjusted TOST $p < .05$ with a stated margin of ± 1 percentage point. A non-significant difference is not equivalence. A small effect can differ significantly from zero while also falling within the equivalence margin; these conclusions answer different questions. Exact sampled identity is reported separately and can arise from structurally inactive factors. The practical-win convention additionally requires at least one percentage point of mean improvement. Inference

and equivalence remain conditional on this simulator and declared margin; neither constitutes equivalence of deployed systems.

VI. RESULTS

A. Strong-Comparator Results

All 22400 evaluation runs complete, including the highest dispatch-cost setting. The current comparisons therefore do not derive their effects from horizon censoring.

Table I leads with the separately tuned comparator. Compute-only contrasts are retained in the supplement; an advantage over compute-only shrinking does not establish an advantage over single-task or retuned fixed-size dispatch.

Core scenarios: the full policy has no practical win over the tuned comparator. It is slower in weak links, churn, combined. Equivalence within the stated margin is supported in uniform.

Activation scenarios: the full policy has no practical win over the tuned comparator. It is slower in prefetch departure. Equivalence within the stated margin is supported in independent RTT, continuous RSSI.

B. Dispatch-Cost Crossover

Fixed-size calibration is repeated independently at every service cost. Table II and Fig. 2 separate a comparison with single-task dispatch from the stronger retuned comparator. The crossover refers to the tested grid; it is not an interpolated universal threshold.

The first tested cost with an adjusted practical win over single-task dispatch is 5000 ms; the preceding grid point is 1000 ms. This brackets an observed transition on the chosen workload, not a physical deployment threshold.

The retuned fixed policy first meets the adjusted practical-win criterion over single-task dispatch at 5000 ms, following 1000 ms on the tested grid. This direct contrast tests amortization without crediting the full policy with its effect.

At 5000 ms, the full policy improves over retuned fixed chunks by 1.26% (simultaneous interval [1.01, 1.52]%). At 20000 ms its signed gain against retuned chunks is -32.10%. A large advantage over single-task dispatch at that point is therefore an amortization result, not superiority over tuned batching.

C. Mechanism Activation

Exact sampled identities, active allocation changes, equivalence, and inconclusive effects are distinct. A mechanism may alter allocation without changing the critical completion path; a non-significant difference alone is not evidence of equivalence.

The independent-delay intervention changes allocations in all 300 worlds, yet the full-versus-no-RTT gain is -0.12% (simultaneous interval [-0.26, 0.02]%). Equivalence within the declared margin is supported. Activation of the clipped scaling regime does not establish useful delay-based ranking.

Continuous eligible-link variation changes allocations in all 300 worlds without eligibility crossings. Its full-versus-no-RSSI gain is 0.04%, with equivalence supported. This

separates an active continuous factor from a practically demonstrated benefit. Exact RTT and all-worker identities in this stationary cell reflect their inactive conditions.

In the compute-heavy departure cell, prefetch has signed gain -3.11% against no prefetch, while the cutoff gains 1.33% against no cutoff. Thus prefetch hurts overall despite a useful cutoff in this cell. The stationary restriction changes allocations in all 300 worlds and suppresses dynamic prefetch, but its gain of 0.59% has simultaneous interval [-0.54, 1.72]%; neither a practical win nor equivalence is established. All mechanism contrasts, including no-prefetch, remain visible in Fig. 3 and the supplement.

VII. INTERPRETATION AND THREATS TO VALIDITY

The RSSI factor can respond to threshold-flapping eligibility, whereas the RTT factor saturates under unloaded echo in the simple core cases. Independent injected delay explicitly exercises its active regime; it is not a measured load-dependent queueing model. RSSI effects in threshold-flapping cases may reflect invalidation risk. The continuous-link cell isolates useful-link variation without crossing eligibility, so allocation changes there cannot be attributed to synthetic departure alone. The result should be read as a mechanism study. Shrinking, link weighting, prefetch, participation restrictions, and session cleanup cannot each be credited with the performance of their combination. Ties and improvements from removals constrain any recommendation. Link-factor inactivity under low conditional RTT is expected from clipping; a negligible gain against single-task dispatch at small costs limits claims of extra scheduler value. Higher service costs can justify batching without establishing that a particular shrinking policy beats retuned fixed chunks.

Calibration remains limited to stationary UDP downlink and conditional unloaded echo. Adapting such curves to changing synthetic signals assumes quasi-stationarity. Equal-airtime allocation, bidirectional behavior, queueing, probe/progress costs, hardware thermals, and return-session transport are not independently calibrated. The fixed RSSI cutoff may reject a usable radio link and is not proof that a person walked away. The main signal/availability scenarios do not constitute a validated physical mobility model.

The study lacks a measured testbed and a faithful close mobile-framework competitor. It cannot establish deployment effectiveness, superiority over published systems, or independent algorithmic novelty. A next validation should use synchronized transfer/probe/task traces and repeated stationary, weak-link, departure/return, and combined trials, holding workloads and recovery semantics fixed across scheduling variants. Runtime control and progress overhead must be included. These limitations constrain the application of the results rather than being replaced by source-level tests.

VIII. CONCLUSION

The study exposes the interaction between dispatch-service cost, allocation granularity, and mechanism activation. Retuning at every cost prevents attributing an advantage over

TABLE I

PRIMARY CONTRASTS AGAINST A CALIBRATION-SELECTED FIXED-SIZE COMPARATOR. GAIN IS MEAN PAIRED RELATIVE IMPROVEMENT; INTERVALS ARE SIMULTANEOUS. EQUIVALENCE USES ADJUSTED TOST WITH A $\pm 1\%$ MARGIN.

Scenario	Fixed tasks	Gain (%)	Simultaneous CI (%)	Difference p	Equivalence p	Complete
uniform	1	0.08	[-0.06, 0.22]	1.0000	< 0.0001	1.000/1.000
weak links	1	-3.67	[-4.15, -3.19]	< 0.0001	1.0000	1.000/1.000
churn	1	-1.52	[-1.81, -1.23]	< 0.0001	1.0000	1.000/1.000
combined	1	-4.65	[-5.34, -3.97]	< 0.0001	1.0000	1.000/1.000
independent RTT	1	0.21	[0.09, 0.34]	< 0.0001	< 0.0001	1.000/1.000
continuous RSSI	1	0.46	[0.33, 0.58]	< 0.0001	< 0.0001	1.000/1.000
prefetch departure	2	-3.08	[-4.54, -1.62]	< 0.0001	1.0000	1.000/1.000

TABLE II

COST SWEEP. THE CONTROL BUDGET IS FIXED AT 64 KB PER ALLOCATION. SERVICE COSTS ARE UNMEASURED ASSUMPTIONS.

Service (ms)	Selected tasks	Full versus single (%)	Simultaneous CI (%)	Full versus tuned (%)	Tuned versus single (%)
0	1	0.25	[0.08, 0.43]	0.25	0.00
1	1	0.26	[0.08, 0.44]	0.26	0.00
10	1	0.26	[0.11, 0.42]	0.26	0.00
100	1	0.32	[0.16, 0.48]	0.32	0.00
1000	1	0.39	[0.23, 0.56]	0.39	0.00
5000	2	3.81	[3.48, 4.15]	1.26	2.58
20000	4	64.94	[64.82, 65.05]	-32.10	73.45

TABLE III

ACTIVATION AUDIT OVER 300 WORLDS PER CELL. ALLOCATION CHANGES COMPARE FULL-POLICY ASSIGNMENT HASHES WITH ABLATIONS. DYNAMIC-PREFETCH COUNTS DIRECTLY EXERCISE THE STATIONARY RESTRICTION.

Scenario	No-RSSI changed worlds	No-RTT changed worlds	Full dynamic prefetch	All-worker dynamic prefetch
continuous RSSI	300	0	0.00	0.00
churn	0	0	0.00	13.98
combined	300	0	0.00	18.88
uniform	0	0	0.00	0.00
weak links	300	0	0.00	0.00
independent RTT	0	300	0.00	0.00
prefetch departure	78	0	0.00	17.50

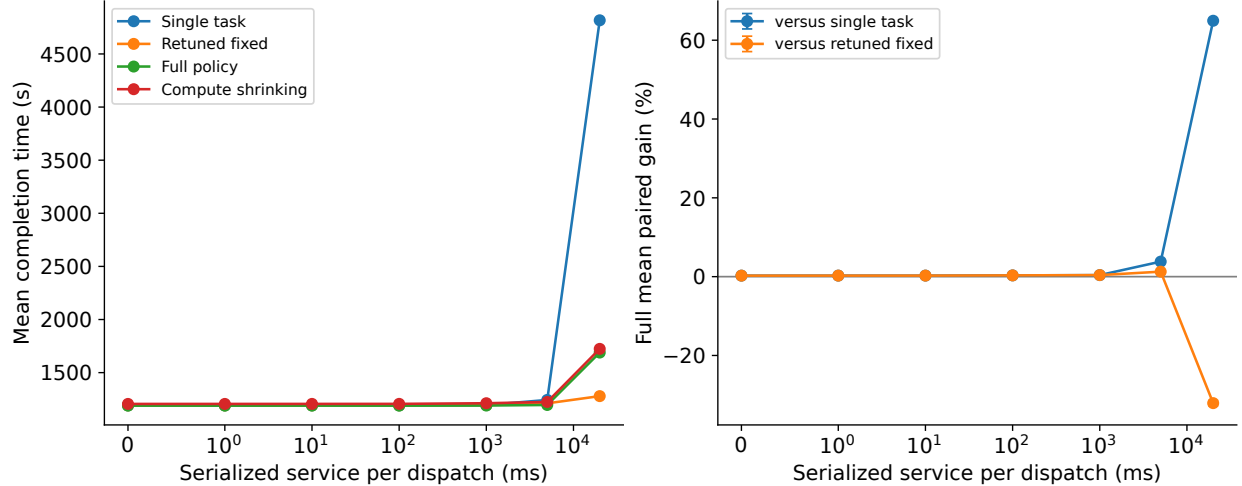


Fig. 2. Dispatch-cost sensitivity with separate calibration at every point. Left: mean completion; right: mean paired relative effects with simultaneous intervals. The zero-cost point still includes dispatch-control traffic. A gain against single-task dispatch must be distinguished from a gain against a retuned comparator.

single-task dispatch to superiority over strong batching. Separate delay, useful-link and departure cells distinguish inactive formula terms from tested performance effects. Common cor-

rected recovery, simultaneous Student inference, and explicit equivalence margins support narrower, reproducible conclusions. Measured coordinator costs and coupled mobile/testbed

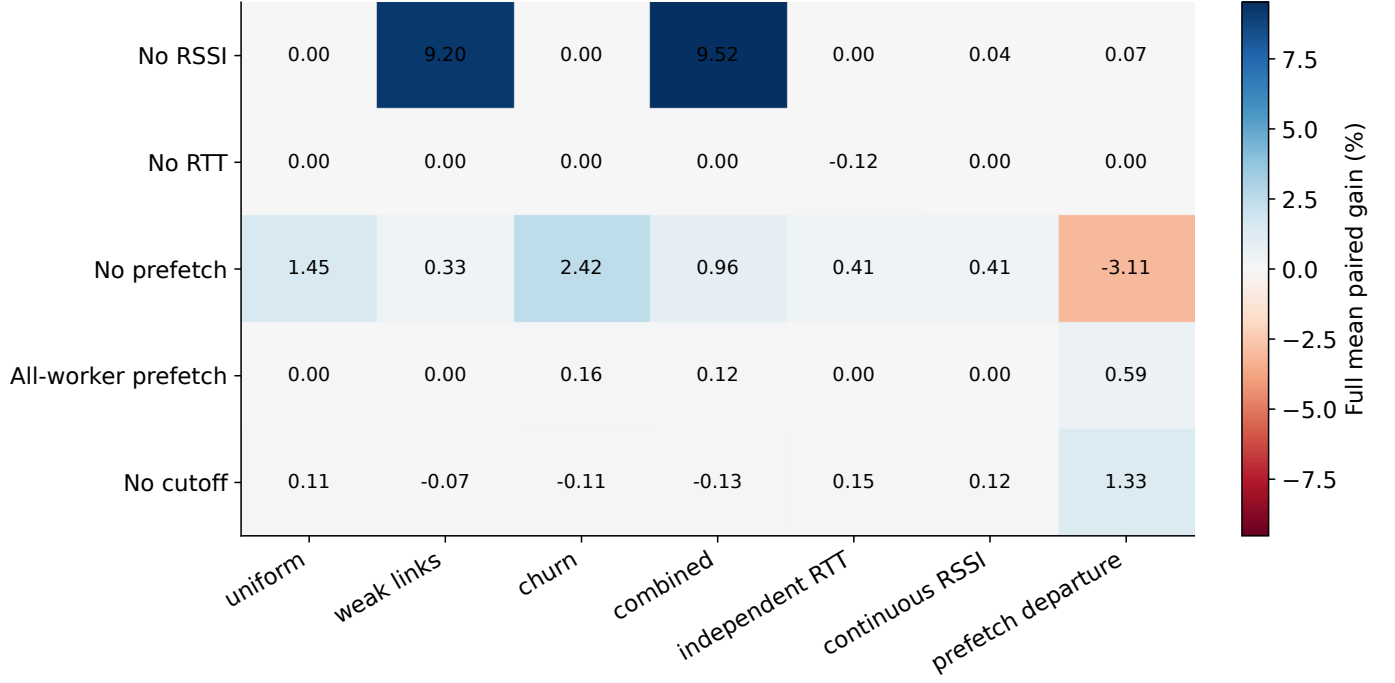


Fig. 3. Mechanism effects across core and activation cells. Color shows signed magnitude, not significance. Negative entries favor the comparator variant. Full statistical and equivalence results, paired distributions, and resource counters appear in the supplement.

validation are required before translating crossover ranges into deployment recommendations.

DATA AND CODE AVAILABILITY

Code, calibration records, all per-run results, analysis scripts, figure sources, and supplementary tables are distributed in the accompanying research artefact at <https://github.com/logiqueviper/HPC/tree/research-deliverables>. The repository provides reproducibility materials; no archival DOI is asserted. The supplementary material lists full policies, calibration knots, uncertainty estimates, completion rates, and recovery counters.

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Anand Kanade His research interests include heterogeneous wireless computing, workload scheduling, adaptive chunk allocation, task recovery, and reproducible simulation.